UNIT - I : INTRODUCTION TO INDUSTRIAL ENGINEERING,

I. Industrial Engineering (I.E.)

Definition

* Industrial Engineering is concerned with the DESIGN, IMPROVEMENT and INSTALLATION of INTEGRATED SYSTEMS of PEOPLE, MATERIALS, INFORMATION, EQUIPMENT and ENERGY. It draws on knowledge of mathematical, physical and social sciences together with the principles of engineering analysis and design to specify, predict and evaluate the results from such systems. (Institute of Industrial Engineers, USA - IISE)

Simple meaning: I.E. = doing the same work with fewer resources - increasing PRODUCTIVITY by eliminating waste of man, material, machine, money and time.

Development of I.E.

Period	Contribution
1776	Adam Smith - 'Wealth of Nations': division of labour
1798	Eli Whitney - interchangeable parts
1832	Charles Babbage - study of time and cost of operations
1880-1911	F. W. Taylor - Scientific Management, time study (father of I.E.)
1900-1912	Frank and Lillian Gilbreth - motion study, therbligs
1910s	Henry Gantt - Gantt chart, task and bonus system; Henry Ford - assembly line
1920s-30s	Elton Mayo - Hawthorne studies; Shewhart - quality control charts
1940s+	Operations research, linear programming, computers, Deming and Juran (TQM)
Today	Lean, Six Sigma, ERP, supply chain, simulation, Industry 4.0

Applications of I.E.

* Manufacturing industries (plant layout, line balancing, production planning and control), service sector (hospitals, banks, airlines, retail), construction, defence, logistics and transport, software and IT, government.
* Typical activities: method study, work measurement, facility location and layout, inventory control, quality control, maintenance planning, cost reduction and value analysis, wage and incentive systems, materials handling, project scheduling (CPM / PERT), safety, ergonomics.

Role of an industrial engineer

* Analyses existing systems and finds the BEST method; designs new facilities / layouts; sets standards of time and cost; improves

productivity and quality; plans manpower, materials and machines; reduces cost and waste; acts as a link between top management and the shop floor (staff function advising line managers); introduces new technology and trains people.

Quantitative tools of I.E.

* Operations Research (linear programming, transportation, assignment, queuing, simulation, network analysis - CPM / PERT), statistics and probability (SQC, forecasting, regression), work study and work measurement, engineering economy, inventory models (EOQ), computer aided design and ERP, decision theory, line balancing, learning curves.

Productivity measurement

* PRODUCTIVITY = relationship between OUTPUT obtained and INPUT used. Increase by producing more with same input or same output with less input.

Productivity = Output / Input

* Partial productivity: labour productivity = units produced / man-hours; machine productivity = output / machine hour; material productivity = output / kg of material. Total productivity = total output value / total input value (labour + material + capital + energy + other).

Type	Formula	Example
Labour	Output / labour hours	500 units / 100 h = 5 units / h
Material	Output / material consumed	500 units / 250 kg = 2 units / kg
Total factor	Output value / total input value	Rs 50,000 / Rs 40,000 = 1.25

* Productivity index = (productivity in current period / productivity in

base period) x 100. Factors affecting: technology, work methods, layout, quality of labour, management, motivation, working conditions, raw material quality.

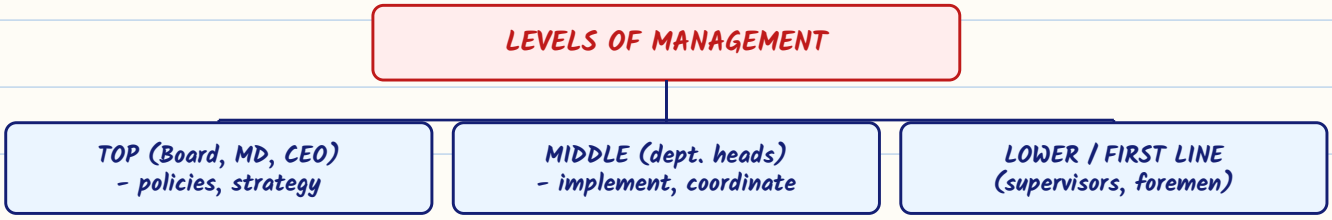
2. Concepts of Management

* MANAGEMENT = the art and science of getting things done through people in formally organised groups, by planning, organising, staffing, directing and controlling resources to achieve organisational goals efficiently and effectively. (Koontz: 'the art of getting things done through and with people in formally organised groups').
* Management is: a PROCESS (series of functions), a SCIENCE (organised knowledge), an ART (skill in application), a PROFESSION, multi-disciplinary, universal, goal oriented, a group activity and continuous.

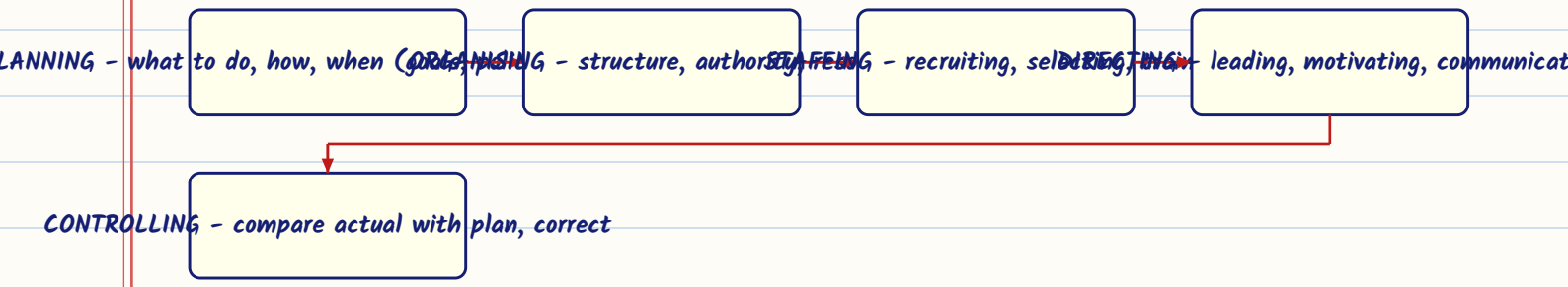
Importance of management

* Achieves group goals; increases efficiency (cuts cost); creates a dynamic organisation which adapts to change; achieves personal objectives; ensures optimum use of resources (5 M's: Men, Machines, Materials, Money, Methods); social benefits - employment, quality goods, welfare; gives stability, growth and competitive advantage.

Levels of management



Functions of management



* PLANNING is the primary function (looking ahead); ORGANISING arranges tasks and relations; STAFFING (HR) puts the right man on the right job; DIRECTING (actuating) guides and motivates; CONTROLLING measures performance against standards and takes corrective action; COORDINATION is the essence of all functions. (POSDCORB - Gulick: Planning, Organising, Staffing, Directing, Coordinating, Reporting, Budgeting).

3. Scientific Management

* Developed by FREDERICK WINSLOW TAYLOR (1856 - 1915, 'Father of Scientific Management'), an American engineer at Midvale and

Bethlehem Steel. Book: 'The Principles of Scientific Management' (1911).

* Definition: 'Knowing exactly what you want men to do and seeing that they do it in the best and cheapest way.' It replaces rule-of-thumb methods by SCIENTIFIC investigation of each task, aiming at maximum output with minimum effort, and prosperity for both employer and employee.

Techniques of Taylor

* Time study, motion study, fatigue study, standardisation of tools and methods, differential piece rate wage system, functional foremanship (8 specialised foremen: 4 in planning - route clerk, instruction card clerk, time and cost clerk, shop disciplinarian; 4 in execution - gang boss, speed boss, repair boss, inspector), scientific selection and training, mental revolution.

Taylor's principles of scientific management

No.	Principle	Meaning
1	Science, not rule of thumb	Study every job scientifically to find the one best way
2	Harmony, not discord	Cooperation between management and workers; no conflict
3	Cooperation, not individualism	Teamwork, mutual help, replacing chaos
4	Maximum output, not restricted output	Maximum production in place of restricted output - prosperity for all
5	Development of each man to his greatest efficiency	Scientific selection, training and development of workers

* Mental revolution: management and workers must change attitude -

both should focus on increasing the surplus (not on dividing it).

* Criticisms: treats workers like machines; ignores human and social needs; monotony, over-specialisation; opposed by trade unions; functional foremanship violates unity of command; applies mostly at shop-floor level.

4. Fayol's Principles of Management

* HENRI FAYOL (1841 - 1925), French mining engineer, 'Father of Modern (Administrative / General) Management'. Book: 'Administration Industrielle et Generale' (1916). Divided industrial activities into: technical, commercial, financial, security, accounting, managerial.

14 principles of management

No.	Principle	Idea
1	Division of work	Specialisation increases efficiency
2	Authority and responsibility	Right to give orders must be matched by responsibility
3	Discipline	Obedience, respect for rules and agreements
4	Unity of command	One employee, ONE boss
5	Unity of direction	One head, one plan for a group of activities with same objective
6	Subordination of individual interest to general interest	Organisation's goal first
7	Remuneration	Fair wages satisfying both employer and employee
8	Centralisation	Degree to which decision making is concentrated at the top
9	Scalar chain	Line of authority from top to bottom (Fayol's gang plank = direct link)
10	Order	A place for everything and everything in its place (material and social order)
11	Equity	Kindliness and justice to employees
12	Stability of tenure of personnel	Avoid unnecessary turnover
13	Initiative	Freedom to think out and execute a plan
14	Esprit de corps	Union is strength - team spirit; harmony

Taylor	Fayol
Scientific management; shop floor level (operational)	General administrative theory; top management level
Focus on efficiency of workers and methods	Focus on managerial functions and principles
Functional foremanship (violates unity of command)	Unity of command (one boss)
Principles developed by observation and experiment	Principles developed from personal experience
Called 'Father of Scientific Management'	Called 'Father of Modern Management'

5. Plant Location and Layout

Factors governing plant location

* Plant location = selecting the geographical site for a factory - a long term decision affecting cost, competitiveness and growth. Two steps: choosing the region, then the exact site.

Factor	Considerations
Nearness to raw materials	Transport cost, availability, quality (steel, cement, sugar plants)
Nearness to market	Perishable / bulky products; distribution cost, customers
Transport facilities	Road, rail, port and air connectivity
Power, water, fuel	Cost and reliability of electricity and water; effluent disposal
Labour supply	Skilled / unskilled availability, wage rates, labour relations
Climate and environment	Humid for textiles, dry for precision instruments
Land, site cost, soil	Price, drainage, future expansion, soil bearing capacity
Government policy	Subsidies, tax holidays, industrial estates, SEZ, pollution control rules
Other	Banking and insurance, housing, schools, hospitals, community attitude, safety, competitors

* Quantitative methods: weighted factor rating, break-even (cost-volume) analysis, transportation model, centre of gravity method (x = sum(w x)/sum(w)).

Centre of gravity: $X = \text{sum}(w_i\, x_i) / \text{sum}(w_i)$, $Y = \text{sum}(w_i\, y_i) / \text{sum}(w_i)$

Plant layout

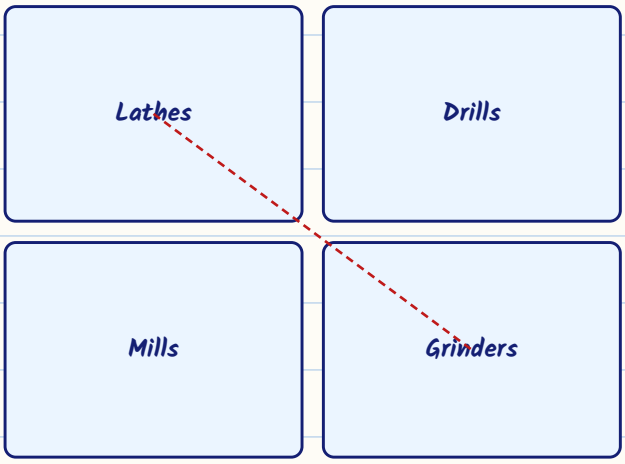
* PLANT LAYOUT = the physical arrangement of machines, equipment, materials, work centres and service areas within a building for the most efficient flow of material and people. Objectives: minimum material handling, smooth flow, minimum production cycle time, efficient use of floor space, safety and comfort, flexibility for future

change.

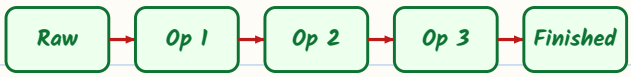
Types of production layouts

* 1. PROCESS (functional) layout - similar machines grouped by function (all lathes together, all drills together). 2. PRODUCT (line) layout - machines arranged in the sequence of operations. 3. FIXED POSITION (static) layout - product stays at a location, men and machines move to it (ships, aircraft, bridges, dams, turbines). 4. COMBINATION (hybrid) layout. 5. GROUP TECHNOLOGY / CELLULAR layout - machines grouped into cells to make a family of similar parts.

PROCESS LAYOUT



PRODUCT (LINE) LAYOUT



Process layout	Product layout
Machines grouped by type / function	Machines in order of operations
General purpose machines, flexible	Special purpose machines, rigid
Low volume, many varieties (job / batch)	High volume, few varieties (mass / continuous)
Skilled labour needed	Semi-skilled labour is enough
High work-in-process, long cycle time	Low work-in-process, short cycle time
More material handling, backtracking	Minimum material handling, straight flow
Breakdown of one machine does not stop all	Breakdown stops the whole line
Low investment per machine, high handling cost	High investment, low unit cost

Advantages and disadvantages

	Process layout	Product layout
Advantages	Flexible, handles variety; lower investment; machine breakdown not critical; better utilisation; job satisfaction, easy supervision of skills	Smooth flow, low handling; low WIP and cycle time; simple control and inspection; less floor space; low unit cost; unskilled labour
Disadvantages	High handling and WIP cost; complex scheduling; long production time; more floor space and inventory; skilled workers needed	Inflexible to product change; heavy investment; one breakdown stops line; monotony; line balancing needed; low volume uneconomical
Applications	Machine shops, hospitals, tool rooms, repair shops, universities	Automobile assembly, bottling, cement, sugar, refineries, TV and appliances

Quantitative techniques for optimal design of layouts

* 1. TRAVEL CHART (from-to chart): matrix of trips / loads between departments; departments with high flow are placed close together. Objective: minimise total material handling cost = sum of (flow x distance x cost per unit distance).
* 2. Relationship (REL) chart (Muther): closeness ratings A (absolutely necessary), E, I, O, U (unimportant), X (undesirable). 3. CRAFT (Computerised Relative Allocation of Facilities Technique) improves an existing layout by exchanging departments; ALDEP, CORELAP construct layouts. 4. Line balancing (assembly line): assign tasks to stations to minimise idle time.

Cycle time C = available time / required output; Min. stations = (sum of

Eg: tasks total 24 min, cycle time 6 min -> minimum stations = 24/6 = 4. If 5 stations are used, efficiency = 24/(5 x 6) = 80%.

* 5. Queueing and simulation models, linear programming and quadratic assignment formulation of layout; Systematic Layout Planning (SLP): flow + activity relationship -> space -> layout.

Revision: IE = design of integrated systems; Taylor - 5 principles (science, harmony, cooperation, max output, development); Fayol - 14 principles; layouts: process, product, fixed position, cellular.